Induction Assignment – The importance of a postgraduate degree in the Computer Science field

Introduction

In the evolving field of Computer Science, and particularly within Cyber Security, two prominent pathways to professional and academic advancement have emerged: postgraduate degrees and industry certifications. While certifications provide targeted, practice-oriented skills that are often immediately applicable in the workplace, postgraduate qualifications such as the Master of Science (MSc) in Cyber Security offer a broader academic foundation, critical thinking skills, and research capabilities. This essay examines the significance of postgraduate education in Cybersecurity. It also compares this academic route to certification-based alternatives. Finally, the essay considers how undertaking a postgraduate degree may influence the field and contribute to society at large.

Grammarly and ChatGPT were used solely for planning, proofreading, and language enhancement. All academic content, reasoning, and conclusions are the author's own.

The Strategic Role of Cyber Security

It is widely acknowledged that Cyber Security has established itself as one of the most rapidly evolving disciplines within the broader field of Computer Science. Evans (2022) asserts, "By 2025, 60% of global IT risk management (ITRM) buyers will depend on risk management solutions to aggregate digital risks in their business … up from 15% in 2019." While the term "risk management solutions" is somewhat vague, it highlights a clear trend, nevertheless. In addition, MarketsandMarkets (2025) projects that the global Cyber Security market size will grow at a compound annual growth rate (CAGR) of 9% through 2030, increasing from USD 227.59 billion in 2025 to USD 351.92 billion. Similarly, according to CompTIA (2024), there is a shortage of around 225,000 professionals to bridge the skill gap in the Cyber Security sector.

There is a growing need to integrate Cyber Security at the strategic level, guiding leadership decisions and long-term planning. Moreover, as noted by Low et al. (2021, cited in Guzmán Sánchez-Mejorada et al., 2021, p. 18), it is essential for educational institutions to continuously update their curricula, particularly concerning soft skills, as this reflects the evolving demand by the industry. Figure 1 (Stavrou and Furnell, 2025) illustrates the distribution of topics within Cyber Security postgraduate programs across the EU and the UK in 2025.

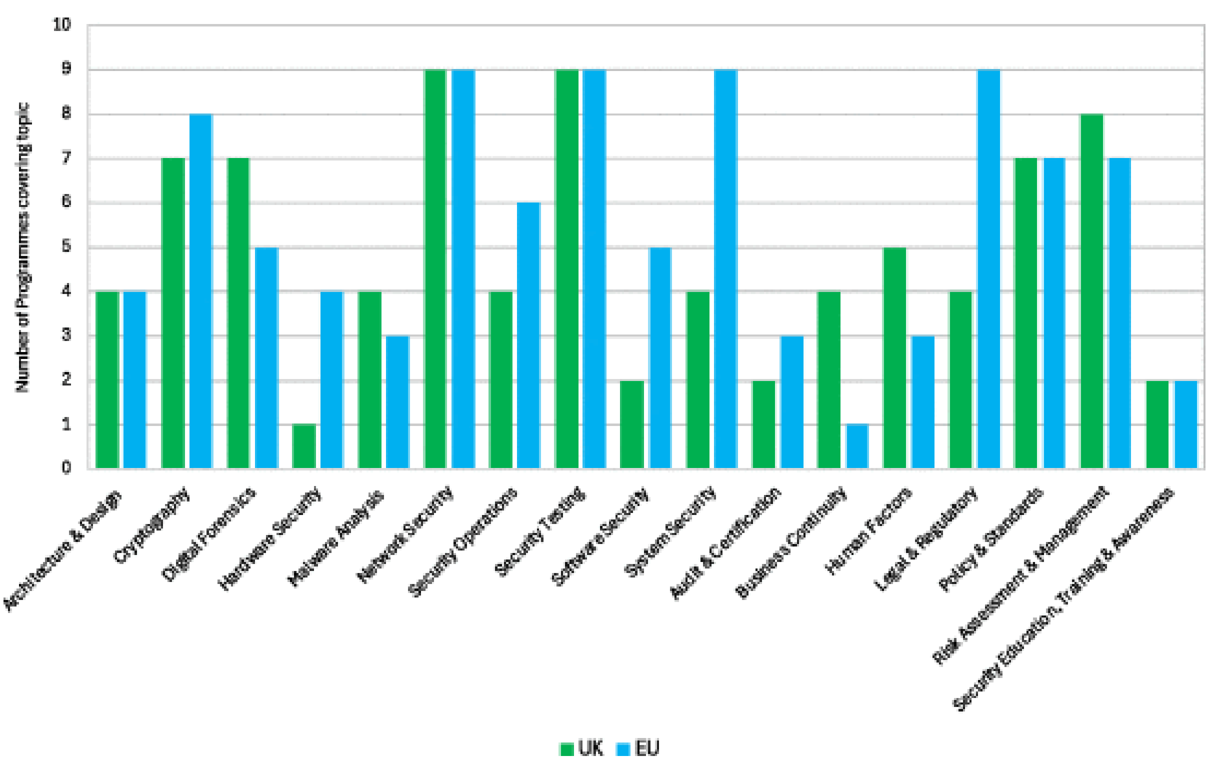

Figure 1. Comparing the topic coverage within the UK and EU sample group.

Source: Stavrou and Furnell (2025, Section IV*)*

The diagram shows that soft skills, such as Human Factors and Security Education, Training & Awareness, remain underrepresented.

Personal experience further illustrates the distinction between practical training and postgraduate degrees.

Personal Reflection from a Cyber Security Analyst

Professional experience as a Cyber Security Analyst has provided practical insight into the technical and operational demands, including incident response, threat detection, and vulnerability assessment. These foundational responsibilities are typically addressed through certification programs and practical training focused on hands-on competence. In this case, Microsoft Security Operations Analyst SC-200 and Implementer Information Protection in Microsoft 365 SC-400 both support operational expertise in cloud-based environments.

Observations from organisational practice indicate that roles lower in the hierarchy tend to focus predominantly on technical execution. In addition, there is a general perception that an MSc may be costly and time-consuming, bringing less value than practical experience, especially for entry-level jobs (American Public University, 2023). In contrast, senior positions such as Chief Information Security Officer (CISO) and Risk and Compliance Manager involve a significantly greater emphasis on strategic planning. While there are certificates that address this, such as the Certified Information Systems Security Professional (CISSP), postgraduate education, such as the MSc in Cyber

Security, enables the development of critical thinking and research skills essential for operating at the strategic level.

Impact on Society and the Cyber Security Field

A postgraduate degree in Cyber Security has the potential to contribute significantly to both the wider society and the field itself. The rising threat to critical infrastructure and public services underscores the need for a workforce with higher education in this area to help shape the foundation for a robust cyber defense strategy at the strategic level. Completion of the MSc in Cyber Security is expected to enhance theoretical understanding in key areas such as Network Security and Security and Risk Management. Furthermore, analytical and research competencies will be developed for addressing emerging cyber threats.

Conclusion

Postgraduate education in Cyber Security plays a vital role in maintaining a strong strategic defense against the constantly growing complexity of the digital threat landscape. While industry certifications provide valuable, specific technical expertise, postgraduate programs offer a broader foundation by combining theoretical knowledge, an analytical and critical thinking mindset, and research skills. As society becomes increasingly dependent on digital infrastructure and services, the contribution of highly educated Cyber Security professionals is important to strengthening the public trust in technology. The MSc in Cyber Security, therefore, represents not only academic achievement but also a strategic investment in digital security within society.

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